

Discrete Structure

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Chapter 1

Set Theory and Logic

- **What is discrete structure?**

- ♣ Fundamental mathematical concepts used in computer science, engineering, and mathematics to represent and analyze discrete (distinct and separate) objects.

Unlike continuous mathematics, which deals with smooth and unbroken values, discrete structures focus on countable elements such as integers, graphs, sets, and logical statements.

- **Why discrete structure?**

- ♣ Enhances problem-solving skills in theoretical and applied computing.

1.1 Law of Set Theory

1. Set - a well-defined collection of objects (Gerge Cantor).

It is an unordered collection of distinct objects. The objects in a set are called the elements, or members, of the set.

2. Set builder form: $\{x : x \text{ is a vowel}\}$

3. Tabular (or listing or roster) form: $\{a, e, i, o, u\}$

* Venn-diagram

4. Descriptive form: The set of vowel letters in English alphabet.

Some special sets of numbers

- $\mathbb{N}$: set of natural numbers.
- $\mathbb{Z}$: set of integers.
- $\mathbb{Q}$: set of rational numbers, $\mathbb{Q}^c = \mathbb{R} \setminus \mathbb{Q}$: set of irrationals.
- $\mathbb{R}$: set of real numbers.
- $\mathbb{C}$: set of complex numbers.

Note: $\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$

Set Operations

If $A, B \subseteq U$ (universal set), then

- $A \cup B = \{x : x \in A \text{ or } x \in B\}$ (union)
- $A \cap B = \{x : x \in A \text{ and } x \in B\}$ (intersection)
- $A - B = \{x : x \in A \text{ and } x \notin B\}$ (set difference)
- A^c (or A') $= U \setminus A = \{x : x \notin A\}$ (set complement)
- $A \Delta B = (A \setminus B) \cup (B \setminus A)$ (symmetric difference)

Types of a set

1. **Null/Empty set** - a set having no elements. It is denoted by $\emptyset$ or $\{\}$.
Example: Set of natural numbers less than 1. Also, $\{\emptyset\}$ is not the empty set. (Think about it.)
2. **Singleton set** - a set having only one element is called singleton set.
Example: Set of even prime numbers.
3. **Finite set** - a set having certain number of elements is called finite set. The number of elements of a finite set A is called **Cardinal number** of A and denoted by $|A|$ or $n(A)$. Example: The set $A = \{1, 2, 3, 4\}$ is finite and $|A| = 4$.
4. **Infinite set** - a set having infinite number of elements is called infinite set.
Example: $\{1, 2, 3, \dots\}$. *Infintie sets can be countable or uncoutable.*
5. **Countable set (discrete set)** - a set having countable number of elements elements. Example: $\{1, 2, 3, \dots\}$. *Every finite set is countable and some infinte sets are countable.*
6. **Uncountable set (continuous set)** - a set whose elements cannot be counted. Example: Set of real numbers between 0 and 1. *Uncountable sets are infinite set.*

Basic terminologies

- ❑ **Universal set:** A fixed set U from which we consider other sets. It is also called universe of discourse.
- ❑ **Intersecting sets:** The two sets A and B are called intersected if $A \cap B \neq \emptyset$.
- ❑ **Disjoint sets:** The two sets A and B are called disjoint if $A \cap B = \emptyset$.
- ❑ **Equivalent sets:** The two sets are called equivalent if they have equal number of elements. Are the sets $A = \{1, 2, 3\}$ and $B = \{a, b, c\}$ equivalent?
- ❑ **Subset:** A set A is said to be a subset of a set B if $x \in A \implies x \in B$. It is denoted by $A \subseteq B$.
Note: *Every set is a subset of itself and Null set is a subset of every set.*

□ **Proper subset:** A subset A of a set B is called proper subset of B if $A \neq B$. Otherwise, it is called improper.

□ **Equal (or identical) sets:** $(A = B) \iff (A \subseteq B) \text{ and } (B \subseteq A)$.

Note: All equal sets are equivalent, but converse is not true.

□ **Power set:** The set of all possible subsets of a set A . It is denoted by $\mathcal{P}(A)$ or 2^A . Example:

If $A = \{a, b, c\}$, then the power set of A is

$$\mathcal{P}(A) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}.$$

Note: The number of possible subsets of a finite set A is $2^{n(A)}$, where $n(A)$ is the *cardinal number* of the set A .

The power set of a set A is the set of all subsets of A , and is denoted by 2^A . That is,

$$2^A = \{S : S \subseteq A\}.$$

For example, for $A = \{2, 4, 17, 23\}$, we have

$$2^A = \{\emptyset, \{2\}, \{4\}, \{17\}, \{23\}, \{2, 4\}, \{2, 17\}, \{2, 23\}, \{4, 17\}, \{4, 23\}, \{17, 23\}, \{2, 4, 17\}, \{2, 4, 23\}, \{2, 17, 23\}, \{4, 17, 23\}, \{2, 4, 17, 23\}\}.$$

The cardinality of this set is 16, and $16 = 2^4$. This is not a coincidence: As we shall see when we get to combinatorics and counting, for a set A with n elements, the cardinality of 2^A is 2^n . This is in fact the reason for the power set notation.

Basic laws of set theory

(a) Idempotent laws	(i) $A \cap A = A$	(ii) $A \cup A = A$
(b) Identity laws	(i) $A \cap U = A$	(ii) $A \cup \emptyset = A$
(c) Domination laws	(i) $A \cup U = U$	(ii) $A \cap \emptyset = \emptyset$
(d) Commutative laws	(i) $A \cap B = B \cap A$	(ii) $A \cup B = B \cup A$
(e) Associative laws	(i) $A \cup (B \cap C) = (A \cup B) \cap C$	(ii) $A \cap (B \cup C) = (A \cap B) \cup C$
(f) Distributive laws	(i) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$	(ii) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
(g) Difference laws	(i) $A - (B \cap C) = (A - B) \cup (A - C)$	(ii) $A - (B \cup C) = (A - B) \cap (A - C)$
(h) Complement laws	(i) $\overline{\overline{A}} = A$	(iv) $\overline{A} = U - A$
	(ii) $\overline{\emptyset} = U$	(v) $\overline{U} = \emptyset$
	(iii) $\overline{A} \cup A = U$	(vi) $\overline{A} \cap A = \emptyset$
(i) De Morgan's law	(i) $\overline{A \cap B} = \overline{A} \cup \overline{B}$	(ii) $\overline{A \cup B} = \overline{A} \cap \overline{B}$
(j) Absorption Laws	(i) $A \cap (A \cup B) = A$	(ii) $A \cup (A \cap B) = A$

Some facts

1. $A \subseteq B \iff (A \cup B = B), (A \cap B = A)$
2. $A \subseteq B \iff \bar{B} \subseteq \bar{A}$
3. $A \cap B = \emptyset \iff (A \subseteq \bar{B}) \text{ and } (B \subseteq \bar{A})$
4. $\bar{A} - \bar{B} = B - A = \bar{A} - B$
5. $(A \subseteq B) \text{ and } (B \subseteq A) \iff (A = B)$
6. $A \Delta B = (A \cup B) - (A \cap B)$
7. $(A \cup B) - C = (A - C) \cup (B - C)$
8. $(A \cup B) - (C - A) = A \cup (B - C)$
9. If $A \cup B = \emptyset$, then $A = \emptyset$ and $B = \emptyset$.
10. $(A \cup B) - C = (A - C) \cup (B - C)$
11. $A \cup (B - C) = (A \cup B) - (C - A)$
12. $(A \cup B) - A = B - A$
13. $A = B \text{ and } B = C \implies A = C$
14. $A - (B - C) = (A - B) \cup (A \cap C)$
15. $A - (B \cup C) = (A - B) - C$
16. $A - (A - B) = A \cap B$
17. $(A - B) - C = (A - C) - B$
18. $A \cap (B - C) = (A \cap B) - (A \cap C)$
19. $A \cap (B \Delta C) = (A \cap B) \Delta (A \cap C)$

Operation rules:

1. $x \in A \cup B \iff x \in A \text{ or } x \in B$
2. $x \in A \cap B \iff x \in A \text{ and } x \in B$
3. $x \in A - B \iff x \in A \text{ and } x \notin B$
4. $x \in \bar{A} \iff x \notin A$
5. $x \notin A \cup B \iff x \notin A \text{ and } x \notin B$
6. $x \notin A \cap B \iff x \notin A \text{ or } x \notin B$
7. $x \notin A - B \iff x \notin A \text{ or } x \in B$
8. $x \notin \bar{A} \iff x \in A$

Example 1.1. Prove that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

$$\begin{aligned}
 x \in A \cup (B \cap C) &\iff x \in A \cup (B \cap C) \\
 &\iff x \in A \text{ or } x \in (B \cap C) \\
 &\iff x \in A \text{ or } (x \in B \text{ and } x \in C) \\
 &\iff (x \in A \text{ or } x \in B) \text{ and } (x \in A \text{ or } x \in C) \\
 &\iff x \in (A \cup B) \text{ and } x \in (A \cup C) \\
 &\iff x \in (A \cup B) \cap (A \cup C) \\
 \therefore A \cup (B \cap C) &= (A \cup B) \cap (A \cup C)
 \end{aligned}$$

Subset: $A \subseteq B : (x \in A \implies x \in B)$

Example 1.2. Prove that

$$(i) \ A - B \subseteq A$$

$$(ii) \ A - B \subseteq \bar{B}$$

Solution:

(i) Let $x \in A - B$ be arbitrary.

$$\implies x \in A \text{ and } x \notin B \text{ (both are true)}$$

$$\implies x \in A \text{ (true)}$$

$$\therefore x \in A - B \implies x \in A$$

Hence $A - B \subseteq A$

(ii) Let $x \in A - B$ be arbitrary.

$$\implies x \in A \text{ and } x \notin B \text{ (both are true)}$$

$$\implies x \notin B \text{ (true)}$$

$$\therefore x \in A - B \implies x \in \bar{B}$$

$$\text{Hence } A - B \subseteq \bar{B}$$

Finite union and intersection

The $\cup$ and $\cap$ notation can be extended to the union and intersection of multiple sets. Given n sets $A_1, A_2, \dots, A_n$, we can write

$$\bigcup_{i=1}^n A_i$$

for their union, and

$$\bigcap_{i=1}^n A_i$$

for their intersection. In fact, this notation is pretty flexible and the same union can be written as

$$\bigcup_{i=1}^n A_i = \bigcup_{1 \leq i \leq n} A_i = \bigcup_{i \in \{x: 1 \leq x \leq n\}} A_i.$$

Here is another example:

$$\bigcap_{i \in \{x: 1 \leq x \leq 10\}} A_i = A_2 \cap A_3 \cap A_5 \cap A_7.$$

Given a set A , the *cardinality* of A , also known as the *size* of A , is simply the number of elements in A . The cardinality of A is denoted by $|A|$. For example, if $A = \{2, 4, 17, 23\}$, then $|A| = 4$.

Sigma and Pi notations

Just as the $\cup$ symbol can be used to compactly express the union of many sets, the $\sum$ symbol can be used to express summations. For example,

$$1 + 2 + \dots + n = \sum_{i=1}^n i = \sum_{1 \leq i \leq n} i = \sum_{i \in \{x: 1 \leq x \leq n\}} i.$$

You should not assume just because $\sum$ appears that there is an actual summation, or that there are any summands at all. For example, when $n = 1$, $\sum_{i=1}^n i = 1$, and when $n \leq 0$, $\sum_{i=1}^n i = 0$!

Similarly, products can be expressed using the $\prod$ symbol, as in

$$2^0 \cdot 2^1 \cdot 2^2 \cdot \dots \cdot 2^n = \prod_{i=0}^n 2^i.$$

One thing to be aware of is that the empty product is defined to equal 1, so

$$\prod_{i=3}^1 i = \prod_{i \in \{2, 4, 10, 14\}} i = 1.$$

A single $\sum$ or $\prod$ symbol can also be used to describe the sum or product over more than one variable. For example,

$$\sum_{1 \leq i, j \leq n} (i + j) = \sum_{i=1}^n \sum_{j=1}^n (i + j).$$

Paradox

♣ **Russell set:** Let R be the set of all sets that are not members of themselves.

If R is not a member of itself, then its definition entails that it is a member of itself; yet, if it is a member of itself, then it is not a member of itself, since it is the set of all sets that are not members of themselves. The resulting contradiction is Russell's paradox.

$$R \in R \iff R \notin R$$

♣ A town has a barber who shaves all men who do not shave themselves. The question is: does the barber shave himself?

If he does, then he is one of the men who shaves himself, contradicting the rule that the barber shaves only those who don't shave themselves. If he doesn't, then he is one of the men who doesn't shave himself, meaning the barber should shave him, again contradicting the rule.

Assignment

- Write the set $\{x : x \in \mathbb{R} \text{ and } x^2 - 4 = 0\}$ in roster form.
- Write the set $\{1, 4, 9, 16, 25, \dots\}$ in set builder form.
- Write the set of real numbers between 0 and 1 in listing form.
- Define power set of a set. Find the power set of $A = \{2, 4, 6\}$.
- Define union, intersection, difference and symmetric difference of two sets.
- If $A = \{1, 3, 5, 7, 9, 11\}$ and $B = \{1, 2, 3, 13\}$, the find $A \cup B, A \cap B, A \Delta B$.
- Let $U = \{x : 1 \leq x \leq 20, x \in \mathbb{N}\}$, $A = \{x : x \text{ is an odd number, } x \in U\}$,
 $B = \{x : x \text{ is a prime number, } x \in U\}$ and $C = \{x : x \text{ is an odd prime number, } x \in U\}$.
 Verify that (i) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ (ii) $(A - B) \cup (B - A) = (A \cup B) - (A \cap B)$.

8. If A, B and C be any sets, prove that

(a) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

(g) $(A \cup B) - (A \cap B) = (A - B) \cup (B - A)$

(b) $\overline{A - B} = B - A$

(h) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

(c) $\overline{A \cup B} = \bar{A} \cap \bar{B}$

(i) $(A \cap B) \cup (A - B) = A$

(d) $A - (B \cup C) = (A - B) \cap (A - C)$

(j) $(A - B) \cup B = A \cup B$

(e) $A - (B \cap C) = (A - B) \cup (A - C)$

(k) $(A - B) \cap B = \emptyset$

(f) $(A \cup B) - C = (A - C) \cup (B - C)$

(l) $A - (B - C) = (A - B) \cup (A \cap C)$

9. Let A and B be any three non-empty subsets of a universal set U , prove that

$$A \Delta B = (A \cup B) - (A \cap B)$$

10. If A, B be any sets, prove that

(a) If $A \cap B = \emptyset$, then $B \subseteq \bar{A}$.

(b) $A \subseteq B \implies \bar{B} \subseteq \bar{A}$

(c) $A - B \subseteq \bar{B}$

(d) $B - A \subseteq B$

(e) $A \subseteq B$ and $B \subseteq C \implies A \subseteq C$

(f) $A \subseteq B \implies \bar{B} \subseteq \bar{A}$

(g) $A - B \subseteq A \cup B$

1.2 Propositions and Logical Operations

- ✓ **Propositional Logic:** Propositional logic is the simplest form of logic. Here the only statements that are considered are propositions, which contain no variables. Because propositions contain no variables, they are either always true or always false (but not both). For example, $3 - 1 = 5$ and $2 + 2 = 4$ propositions because $3 - 1 = 5$ is always false and $2 + 2 = 4$ is always true. But $x + 3 = 7$ is not proposition because it depends on the value of x .

Whether a statement is a proposition or not is a structural property. But if a statement doesn't contain any variables (or other undefined terms), it is a proposition, and as a side-effect of being a proposition it's always true or always false.

- ✓ **Truth value:** The truthfulness or falsity of a given statement is called truth value of the statement. $True = T$ and $False = F$.
- ✓ **Negation of a statement:** The statement which denies the given statement p is called negation statement of p . It is denoted by $\neg p$.
- ✓ **Compound statement:** A combination of two or more statements is called compound statement.

1.2.1 Compound statements with logical connectives

Let p, q be any two statements.

□ **Conjunction:** p AND q .

$$p \wedge q = \begin{cases} T & \text{if both } T \\ F & \text{otherwise} \end{cases}$$

□ **Disjunction:** p OR q .

$$p \vee q = \begin{cases} F & \text{if both } F \\ T & \text{otherwise} \end{cases}$$

□ **Conditional statement (implication):**

IF p , THEN q .

$$p \implies q = \begin{cases} F & \text{if } p = T, q = F \\ T & \text{otherwise} \end{cases}$$

□ **Biconditional statement:** p IF AND ONLY IF q .

$$p \iff q = \begin{cases} T & \text{if both same} \\ F & \text{otherwise} \end{cases}$$

Exclusive OR. If we want to exclude the possibility that both p and q are true, you can use EXCLUSIVE OR instead. This is written as $p \oplus q$, and

p	q	$p \wedge q$	$p \vee q$	$p \implies q$	$p \iff q$
T	T	T	T	T	T
T	F	F	T	F	F
F	T	F	T	T	F
F	F	F	F	T	T

Table 1.1: Formulae (truth table) for logical connectives

Example 1.3. Translate the following statement into logical form.

"You can access the internet from campus if and only if you are a computer science major or you are not a freshman."

- p : You can access the internet from campus.
- q : You are a computer science major.
- r : You are a freshman.

Logical form: $p \leftrightarrow (q \vee \neg r)$

Example 1.4. Translate the following statement into logical form.

"The automated reply cannot be sent when the file system is full."

- p : The file-system is full
- q : The automated reply can be sent

Logical form: $p \rightarrow \neg q$

Better to know conditional statement (If p , then q .)

- q if p .
- q whenever p .
- q , provided that p .
- Whenever p , then also q .
- p is a sufficient condition for q .
- For q , it is sufficient that p .
- q is a necessary condition for p .
- For p , it is necessary that q .
- p only if q .

Note: p is called antecedent (hypothesis) and q is called consequent (result or conclusion).

Examples

- If a number is even, then it is divisible by 2.
- You cannot enter unless you have a ticket.
- A triangle is equilateral only if all angles are 60° .
- When water reaches 100°C , it boils.
- No shoes, no service.

Converse, Inverse and Contrapositive statements

Consider a conditional statement $p \implies q$, where p and q be two simple statements. Then

Converse	Inverse	Contrapositive
$q \implies p$	$\neg p \implies \neg q$	$\neg q \implies \neg p$

Example 1.5. Consider $p \implies q$: ‘If two lines are perpendicular, then they are not parallel.’ Then p : ‘Two lines are perpendicular.’ and q : ‘Two lines are not parallel.’

* *Converse* ($q \implies p$): ‘If two lines are not parallel, then they are perpendicular.’

* *Inverse* ($\neg p \implies \neg q$): ‘If two lines are not perpendicular, then they are parallel.’

* *Contrapositive* ($\neg q \implies \neg p$): ‘If two lines are parallel, then they are not perpendicular.’

Conclusion:

- A conditional statement and its converse are not logically equivalent.
- A conditional statement and its inverse are not logically equivalent.
- A conditional statement and its contrapositive statement are logically equivalent.

Equivalent propositions

The two propositions p and q are equivalent (denoted by $p \equiv q$) if they have same truth values.

Note: Biconditional proposition $p \iff q$ is equivalent to the conjunction of $p \implies q$ and its converse proposition.

$$\text{i.e., } p \iff q \equiv (p \implies q) \wedge (q \implies p)$$

[In this case, we say that p and q are necessary and sufficient conditions for each other.]

Tautology

A compound proposition which is always true whatever the truth value of its components.

For example, $p \vee (\neg p)$ is a tautology.

Contradiction

A compound proposition which is always false whatever its components’ truth values.

For example, $p \wedge (\neg p)$ is a contradiction.

Contingency

A compound proposition that is neither tautology nor contradiction. For example, the statement $(p \wedge \neg q) \rightarrow (p \wedge q)$ is contingency.

1.2.2 Equivalent laws

Let p, q, r be any statements, t stands for tautology and c for contradiction.

(a) Idempotent laws:	(i) $p \wedge p \equiv p$	(ii) $p \vee p \equiv p$
(b) Commutative laws:	(i) $p \wedge q \equiv q \wedge p$	(ii) $p \vee q \equiv q \vee p$
(c) Identity laws:	(i) $p \wedge t \equiv p$	(ii) $p \vee c \equiv p$
(d) Domination laws:	(i) $p \wedge c \equiv c$	(ii) $p \vee t \equiv t$
(e) Associative laws:	(i) $p \vee (q \vee r) \equiv (p \vee q) \vee r$	(ii) $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
(f) Distributive laws:	(i) $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	(ii) $p \wedge (p \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
(g) De Morgan's laws:	(i) $\neg(p \wedge q) \equiv \neg p \vee \neg q$	(ii) $\neg(p \vee q) \equiv \neg p \wedge \neg q$
(h) Complement laws:	$\neg(\neg p) \equiv p$	
(i) Contrapositive law:	$[p \implies q] \equiv [(\neg q) \implies (\neg p)]$	
(j) Law of syllogism:	$(p \implies q) \wedge (q \implies r) \implies (p \implies r)$	

Tautologies

- $(p \wedge q) \implies p$
- $p \implies (p \vee q)$
- $\neg p \implies (p \implies q)$
- $(p \wedge (p \implies q)) \implies q$
- $(\neg q \wedge (p \implies q)) \implies \neg p$
- $(p \wedge q) \implies q$
- $q \implies (p \vee q)$
- $q \implies (p \vee q)$
- $\neg(p \implies q) \implies p$
- $(\neg p \wedge (p \vee q)) \implies q$

Examples of proving equivalent statements

Example 1.6. Prove that $p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$

$$\begin{aligned}
 (p \leftrightarrow q) &\equiv (p \rightarrow q) \wedge (q \rightarrow p) \\
 &\equiv (\neg p \vee q) \wedge (\neg q \vee p) \\
 &\equiv (p \wedge q) \vee (\neg p \wedge \neg q) \\
 &\equiv [(\neg p \wedge \neg q) \vee (p \wedge q)] \\
 &\equiv [(p \wedge q) \vee (\neg p \wedge \neg q)]
 \end{aligned}$$

Example 1.7. Verify the logical equivalence: $\neg(\neg p \wedge q) \wedge (p \vee q) \equiv p$ with reasons.

Solution

$$\begin{aligned}
 &\neg(\neg p \wedge q) \wedge (p \vee q) \\
 &\equiv (\neg(\neg p) \vee \neg q) \wedge (p \vee q) && \text{(De Morgan's Law)} \\
 &\equiv (p \vee \neg q) \wedge (p \vee q) && \text{(Double Negation)} \\
 &\equiv p \vee (\neg q \wedge q) && \text{(Distributive Law)} \\
 &\equiv p \vee \mathbf{c} && \text{(Negation Law, where } \mathbf{c} \text{ is a contradiction)} \\
 &\equiv p && \text{(Identity Law)}
 \end{aligned}$$

This can also be proven by constructing a truth table.

p	q	$\neg p$	$\neg p \wedge q$	$\neg(\neg p \wedge q)$	$p \vee q$	$\neg(\neg p \wedge q) \wedge (p \vee q)$
T	T	F	F	T	T	T
T	F	F	F	T	T	T
F	T	T	T	F	T	F
F	F	T	F	T	F	F

Table 1.2: First and Last column have same truth values

Negation laws:

- | | | |
|---|--|---|
| 1. $\neg(\neg p) \equiv p$
2. $\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$ | | 3. $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$
4. $\neg(p \implies q) \equiv p \wedge (\neg q)$ |
|---|--|---|

DON'T FORGET !

	p	$\neg p$
(i)	All (every)	Some ... not ...
(ii)	Some (at least one)	None

1.2.3 Predicates and Quantifiers

- Universal quantifier: $\forall$ - for every or for all (or *something equivalent*).
- Existential quantifier: $\exists$ - there exists or there is at least one (or *something equivalent*).

Definition 1.8 (Predicate). The set builder notation

$$\{x \mid p(x)\}$$

is just the collection of all objects x for which p is sensible and true. Such a sentence $p(x)$ is called a **predicate**.

$p(x)$ is also called a **propositional function** because each choice of x produces a proposition $p(x)$ that is either true or false.

Predicates in universal quantification

The universal quantification of a predicate $p(x)$ is the statement: "For all values of x , $p(x)$ is true."

The universal quantification of $p(x)$ is denoted by

$$\forall x p(x).$$

Example 1.9. The sentence $p(x) : -(-x) = x$ is a predicate that makes sense for real numbers x . The universal quantification of $p(x)$, $\forall x p(x)$, is a true statement, because for all real numbers:

$$-(-x) = x.$$

Example 1.10. Let $q(x) : x + 1 < 4$. Then $\forall x \geq 0$, $q(x)$ is a false statement, because $q(5)$ is not true.

Predicates in existential quantification

The existential quantification of a predicate $p(x)$ is the statement: "There exists a value of x for which $p(x)$ is true."

The existential quantification of $p(x)$ is denoted by:

$$\exists x p(x).$$

Example 1.11. For $q(x) : x + 1 < 4$, the existential quantification $\exists x q(x)$ is a true statement because $q(2)$ is true.

Example 1.12. The statement $\exists y (y + 2 = y)$ is false. There is no value of y for which the propositional function $y + 2 = y$ produces a true statement.

Practice Questions

Find the truth value of the following statements:

(a) $(\forall x)(\exists y)(x + y < 3)$

(b) $(\exists y)(\forall x)(x + y < 3)$

(c) $(\exists y)(\forall x)(x + y = 3)$

Quantifier Negation and Distribution

Example 1.13. Show that $\neg(\forall x p(x)) \equiv \exists x \neg p(x)$.

Solution

$\forall x p(x)$ is False if there is at least one x for which $p(x)$ is False: in other words, if $\exists x \neg p(x)$ is True. Thus, we can express this relationship in symbolic form as:

$$\neg(\forall x p(x)) \equiv \exists x \neg p(x).$$

Example 1.14. Show that $\exists x (p(x) \vee q(x)) \equiv \exists x p(x) \vee \exists x q(x)$.

Solution

The statement $\exists x (p(x) \vee q(x))$ is True if and only if there exists an x such that either $p(x)$ or $q(x)$ is true. This means either:

$$\exists x p(x) \quad \text{or} \quad \exists x q(x)$$

is True. Therefore, it holds if and only if:

$$\exists x p(x) \vee \exists x q(x)$$

is True.

1.3 Methods of Proof

To begin the idea, let us construct the truth table of the compound statement

$$(p \wedge (p \implies q)) \implies q.$$

	p	q	$p \implies q$	$p \wedge (p \implies q)$	$(p \wedge (p \implies q)) \implies q$
1	T	T	T	T	T
2	T	F	F	F	T
3	F	T	T	F	T
4	F	F	T	F	T

The statement $(p \wedge (p \implies q)) \implies q$ is a **tautology**. Therefore, this implication is true regardless of the truth values of its components p and $p \implies q$. In this case, we say that q **logically follows** from p and $p \implies q$. This we can write as:

$$\begin{array}{c}
 p \\
 p \implies q \\
 \hline
 \therefore q
 \end{array}$$

1.3.1 Logical Arguments

In general, if a statement q follows from the statements $p_1, p_2, \dots, p_n$, that is, the implication

$$(p_1 \wedge p_2 \wedge \dots \wedge p_n) \implies q$$

is a tautology, we write:

$$\begin{array}{c}
 p_1 \\
 p_2 \\
 \vdots \\
 \hline
 p_n \\
 \therefore q
 \end{array}$$

This means that if we know that p_1 is true, p_2 is true, $\dots$, p_n is true, then we know q is true. The p_i 's are called the **hypotheses** or **premises**, and q is called the **conclusion**.

The above assertion is called an **argument**, that is, by an argument, we mean a sequence of statements that end with a conclusion.

Definition 1.15 (Argument or inference). An **argument** or **inference** is an assertion (logic of reasoning) that a given set of propositions $p_1, p_2, \dots, p_n$ called hypotheses or premises yield another proposition q called the conclusion.

Definition 1.16 (Valid argument). An argument $p_1 \wedge p_2 \wedge \dots \wedge p_n \implies q$ is said to be **valid** if the statement

$$p_1 \wedge p_2 \wedge \dots \wedge p_n \implies q$$

is a tautology.

Thus we can say that the argument

$$p_1 \wedge p_2 \wedge \dots \wedge p_n \implies q$$

is valid if q is true whenever all the premises $p_1, p_2, \dots, p_n$ are true.

1.3.2 Argument Validity

Note: If any one of $p_1, p_2, \dots, p_n$ is false, then the hypothesis $p_1 \wedge p_2 \wedge \dots \wedge p_n$ is false and the implication $p_1 \wedge p_2 \wedge \dots \wedge p_n \implies q$ is automatically true, regardless of the truth value of q .

An argument which is not valid is called a **fallacy**.

It is to be noted that the validity of an argument depends only on the form of the statements involved and not on the truth values of the variables that they contain. Such arguments are called the **rules of inference**.

Important Rule of Inference

A very important rule of inference is:

$$\frac{p \quad p \implies q}{\therefore q}$$

1.3.3 Rules of Inference

In general, any tautology of the form $p \rightarrow q$ can be used to “manufacture” the inference rule $p \rightarrow q$ that says that if we know p we can conclude q . Similarly, a tautology of the form $p \leftrightarrow q$ can be converted into the equivalence $p \leftrightarrow q$, which can be regarded as two inference rules, $p \rightarrow q$ and $q \rightarrow p$. A particularly important inference rule is called *modus ponens*, and says that if we know that p and $p \rightarrow q$ are both true, we can conclude that q is true. It follows from the tautology $(p \wedge (p \rightarrow q)) \rightarrow q$:

p	q	$p \rightarrow q$	$p \wedge (p \rightarrow q)$	$(p \wedge (p \rightarrow q)) \rightarrow q$
T	T	T	T	T
T	F	F	F	T
F	T	T	F	T
F	F	T	F	T

We’ve already seen a number of inference rules above, like $(p \rightarrow q) \leftrightarrow (\neg q \rightarrow \neg p)$, without calling them that. Here are three others, all corresponding to tautologies that you are invited to verify using truth tables:

$$(\neg p \rightarrow \{ \}) \leftrightarrow p$$

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (q \rightarrow p)$$

$$(p \leftrightarrow q) \leftrightarrow (p \rightarrow q) \wedge (\neg p \rightarrow \neg q)$$

These three rules are of particular importance. The first formally establishes the validity of proofs by contradiction, and the second and third provide two means for proving “if and only if” statements. We’ve been using these all along, but now we know why they are justified.

Modus Ponens (Rule of Detachment)

This rule of inference is well known as **Modus Ponens** or **Rule of Detachment** (from Latin "modus ponendo ponens" meaning "the method of affirming"). It is universally valid:

$$\frac{p \quad p \implies q}{\therefore q}$$

Law of Syllogism

Another important rule of inference is the **Law of Syllogism**:

$$\frac{p \implies q \quad q \implies r}{\therefore p \implies r}$$

This rule is universally valid (verify by showing that the conjunction $(p \implies q) \wedge (q \implies r) \implies (p \implies r)$ is a tautology).

Modus Tollens

The **Modus Tollens** rule (from Latin "modus tollendo tollens" meaning "the method of denying") is given by:

$$\frac{p \implies q \quad \sim q}{\therefore \sim p}$$

Example 1.17. Is the following argument valid?

- Smoking is healthy.
 - If smoking is healthy, then cigarettes are prescribed by physicians.
- $\therefore$ Cigarettes are prescribed by physicians.

Solution

The argument is valid since it is of the form **Modus Ponens**. However, the conclusion is **false**. Observe that:

- The first premise p : "smoking is healthy" is **false**.
- The second premise $p \implies q$ is then **true** (a false antecedent makes the implication true).
- The conjunction $(p \wedge (p \implies q))$ is **false** (since p is false).

Example 1.18. Is the following argument valid?

- If you invest in the stock market, then you will get rich.
 - If you get rich, then you will be happy.
- $\therefore$ If you invest in the stock market, then you will be happy.

Solution

The argument is valid since it is of the form **Law of Syllogism**. Although the conclusion may be false in reality, the logical form is correct.

$$\begin{array}{c} p \implies q \\ q \implies r \\ \hline \therefore p \implies r \end{array}$$

where:

p : you invest in the stock market

q : you will get rich

r : you will be happy

Example 1.19. Determine the validity of the following argument:

- If I drive to work, then I will arrive tired.
- I do not drive to work.

Solution

I will not arrive tired.

Then the argument is of the form

p : I drive to work

and

q : I will arrive tired

$$\begin{array}{c} p \implies q \\ \sim p \\ \hline \therefore \sim q \end{array}$$

Now to test the validity of the argument, construct the truth table of the statement $(p \implies q) \wedge (\sim p) \implies (\sim q)$.

Assignment

1. If p : Every prime number is odd and q : 2 is a prime number, express each of the followings in to words.
 - (a) $\neg p \vee \neg q$
 - (b) $\neg(p \wedge q)$
 - (c) $p \implies q$
 - (d) $\neg q \implies \neg p$
2. Let p : A person is intelligent and q : A person is a mathematician. Express each of the followings in symbolic form.
 - (a) A person is intelligent and he is a mathematician.

- (b) A person is neither intelligent nor mathematician.
 (c) A person is a intelligent, provided that he is a mathematician.
 (d) It is false that a person is intelligent implies he is a mathematician.
3. Write the converse, inverse and contrapositive statements to the following conditional statements.
- (a) If a triangle is equilateral, then it is isosceles.
 (b) If a line makes an angle of 45° with positive x-axis, then its slope is 1.
 (c) If a number is divisible by 2 and 3, then it is divisible by 6.
 (d) A number is even if it is divisible by 2.
 (e) If a set is finite, then it is countable.
 (f) If a set is uncountable, then it is infinite.
4. Write the negation to each of the following statement with their truth value.
- | | |
|--|---|
| (a) Every parallelogram is a rhombus. | (h) 2 is the only even prime number. |
| (b) All triangles are equilateral triangle. | (i) $\sqrt{-4}$ is an irrational number. |
| (c) π is an irrational number and $\frac{22}{7}$ is a rational number. | (j) The square of every real number is positive. |
| (d) Some even numbers are prime. | (k) If 3 is prime, then 7 is even. |
| (e) Every set is a subset of itself. | (l) If $2 < 5$ implies that $3 > 8$, then 6 is odd. |
| (f) Null set is a subset of every set. | (m) A function is continuous whenever it is differentiable. |
| (g) The square of a negative number is positive. | (n) 1 is neither prime nor composite. |
5. Write the converse, inverse and contrapositive of the following conditional statements with truth values.
- (a) If two lines are not perpendicular, then they are parallel.
 (b) If two triangles are congruent, then they are similar.
 (c) If x is greater than 1, then x^2 is greater than 1.
6. If p and q are any two statements, prove that
- | | |
|--|--|
| (a) $(p \wedge q) \implies (p \vee q)$ is a tautology. | (d) $(p \wedge q) \wedge \neg(p \vee q)$ is a contradiction. |
| (b) $\neg(p \implies \neg q)$ is a contradiction. | (e) $(p \wedge q) \implies (p \vee q)$ is a tautology. |
| (c) $\neg(\neg p \wedge q) \equiv p \vee \neg q$. | (f) $(\neg p) \wedge (p \wedge q)$ is a contradiction. |
7. If p and q are any two statements, prove that
- | | |
|--|---|
| (i) $(p \implies q) \equiv (\neg q \implies \neg p)$ | (ii) $\neg(\neg p \vee q) \equiv (p \wedge \neg q)$ |
|--|---|

8. If p, q, r are any statements and c stands for contradiction, show that the following statements are true.

$$\begin{array}{ll} \text{(i)} (\neg p \implies c) \iff p & \text{(ii)} (p \vee q) \vee (\neg p \wedge \neg q) \\ \text{(iii)} [(p \wedge q) \implies r] \iff [p \implies (q \implies r)] & \text{(iv)} (\neg p \wedge q) \iff \neg(p \vee \neg q) \end{array}$$

9. Determine the validity of the following argument

I will become famous or I will be a writer.

I will not be a writer.

: I will become famous.

10. Determine the validity of the following argument

If I graduate this semester, then I will have passed the math course.

If I do not study math for 10 hours a week, then I will not pass math.

If I study math for 10 hours a week, then I cannot play football.

: If I play football, I will not graduate this semester.

1.4 Methods of Proof

Definition 1.20 (Theorem). A true statement is known as a **Theorem**, and the reason why it is true is its Proof.

Theorem 1.21. *15 is a multiple of 3.*

Proof. $15 = 5 \times 3$. □

But the statement "20 is a multiple of 3" is false. So the statement is not a Theorem.

Theorem 1.22. *20 is not a multiple of 3.*

Proof. If 20 were a multiple of 3, there would be an $n \in \mathbb{N}$ such that $n \times 3 = 20$. Since $7 \times 3 = 21$, n must be less than 7. But none of the values 1, 2, 3, 4, 5, 6 works. □

Example 1.23. Prove or disprove the statement that if $x, y \in \mathbb{R}$, $(x^2 = y^2) \iff (x = y)$.

Solution: The statement can be restated as

$$(\forall x)(\forall y)(\mathbb{R}(x, y)), \quad (x^2 = y^2) \iff (x = y)$$

Thus, to prove this result, we would need to provide steps, each of which would be true for all $x, y \in \mathbb{R}$. To disprove this result, we need only find one example for which this implication is false. Since $(-3)^2 = 3^2$, but $-3 \neq 3$, the result is false. Our example is called a **counter example** and any other counter example would do just as well.

From the above example, we observe that

- A disprove of a statement is a proof that the statement is false.

- A counter example to a statement of the form $(\forall x \in D)p(x)$ is an element $y \in D$ for which $p(y)$ is false. Thus the method of disprove by counter example is based on the following fact:

$$\sim [(\forall x \in D)p(x)] \iff (\exists x \in D)[\sim p(x)]$$

Meaning to prove the theorem

To prove the theorem means to show that the implication

$$(p_1 \wedge p_2 \wedge \cdots \wedge p_n) \implies q$$

is a tautology.

Note that we are not trying to show that q (the conclusion) is true, but only that q will be true if all the p_i 's are true. For this reason, mathematical proofs often begin with the statement "suppose that $p_1, p_2, \dots, p_n$ are true" and conclude with the statement "therefore q is true". The proof does not show that q is true, but simply shows that q has to be true if the p_i 's are all true.

Proving invalidity of an argument

To prove that an argument or an inference pattern is invalid, produce a counter example to the inference pattern. An example for which an inference/argument is false, is called a **counter example**.

Example 1.24. $n^2 - n + 41$ is a prime for all $n \in \mathbb{N}$.

solution

Counter example: $n = 41$ results in a composite number.

2.1.3 Proving the validity of an argument

To prove that an inference pattern is valid, prove that the conditional

$$\text{Conjunction of premises} \implies \text{Conclusion}$$

is a tautology.

Generally two methods in proving validity:

- (a) Direct proof
- (b) Indirect proof

1.4.1 Direct Method

Direct Method

A direct proof assumes the hypothesis p and uses logical deductions to arrive at the conclusion q . Starting from the true premises (in mathematical proof), the process of reaching the true conclusion is called direct proof. Generally two rules of inference are in use:

(i) **First rule of inference** or *modus ponens*

$$(p \wedge (p \implies q)) \implies q$$

(ii) **Second rule of inference** or *law of syllogism*

$$((p \implies q) \wedge (q \implies r)) \implies (p \implies r)$$

Definition 1.25. An integer n is a perfect square if there is an integer m such that $n = m^2$.

Theorem 1.26. If n is an even integer, then n^2 is even.

Proof. Assume n is even. Then, by definition, $n = 2k$ for some integer k . Squaring both sides:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2).$$

Since $2k^2$ is an integer, n^2 is even. □

Assignment

Prove the following statements.

1. If n is an odd integer, then n^2 is odd.
2. If m and n are both perfect square the mn is also a perfect square.

1.4.2 Indirect method

(i) **Method of contrapositive**

To prove the implication $p \implies q$, prove the implication $\neg q \implies \neg p$. That is, assume q is false, and show that p is then false. It is based on the tautology:

$$(p \implies q) \iff ((\neg q) \implies (\neg p))$$

Theorem 1.27. If n^2 is odd, then n is odd.

Proof. We prove the contrapositive: If n is even, then n^2 is even. Assume n is even. Then $n = 2k$ for some integer k . Thus:

$$n^2 = (2k)^2 = 4k^2 = 2(2k^2),$$

which is even. Hence, the original statement holds. □

(ii) **Method of contradiction**

This method is based on the tautology:

$$((p \implies q) \wedge (\neg q)) \implies (\neg p)$$

Theorem 1.28. $\sqrt{2}$ is irrational.

Proof. Assume, for contradiction, that $\sqrt{2}$ is rational. Then, $\sqrt{2} = \frac{a}{b}$, where a and b are coprime integers. Squaring both sides:

$$2 = \frac{a^2}{b^2} \implies a^2 = 2b^2.$$

Thus, a^2 is even, so a must be even (by the previous theorem). Let $a = 2k$. Substituting:

$$(2k)^2 = 2b^2 \implies 4k^2 = 2b^2 \implies b^2 = 2k^2.$$

This implies b^2 is even, so b is even. But this contradicts the assumption that a and b are coprime. Hence, $\sqrt{2}$ is irrational. $\square$

Strategy of proving a mathematical statement

When we want to prove a statement of the form $\forall x (P(x) \rightarrow Q(x))$:

- First try to prove it using a **direct proof**
- If a direct proof doesn't seem to work, try a **proof by contraposition**

Assignment

1. Prove or disprove the statement $(\forall x, y \in \mathbb{R})(x^2 < y^2 \implies x < y)$.
2. Prove that the sum of two rational numbers rational. [direct]
3. Prove or disprove the statement: every prime number is odd.[counter example]
4. Prove or disprove: $n^2 + 41n + 41$ is a prime number for every integer n .[]
5. Prove that the sum of two even numbers is even.[direct]
6. Prove that the sum of two odd numbers is odd.[]
7. Prove that n^2 is even if and only if n is even.[direct]
8. Prove that $(\forall n \in \mathbb{N})(n^2 + n)$ is even.
9. If n is an integer and $3n + 2$ is odd, then n is odd. [contrapositive]
10. Prove that if $n = ab$ where a and b are positive integers then $a \leq \sqrt{n}, b \leq \sqrt{n}$. [contrapositive]
- 11.

1.4.3 Other forms of proof

Vacuous Proof

$p \rightarrow q$ is true when p is false. If we can show that p is false, then we have a proof called **vacuous proof**.

We can quickly prove that a conditional statement $p \rightarrow q$ is true when we know that p is false, because $p \rightarrow q$ must be true when p is false. Consequently, if we can show that p is false, then we have a proof, called a **vacuous proof**, of the conditional statement $p \rightarrow q$.

Example: Show that the proposition $P(0)$ is true where the domain consists of the integer numbers and $P(n)$ is “If $n \geq 1$ then $n^2 > n$.”

Note that $P(0)$ is “If $0 \geq 1$, then $0^2 > 0$.” We can show $P(0)$ using a vacuous proof, because the hypothesis $0 \geq 1$ is false. Hence $P(0)$ is automatically true.

Trivial Proof

A proof of $p \rightarrow q$ that uses the fact that q is true is called **trivial proof**.

By showing that q is true, it quickly follows that $p \rightarrow q$ must also be true. A proof of $p \rightarrow q$ that uses the fact that q is true is called a **trivial proof**.

Example: Let $P(n)$ be “If a and b are positive integers with $a \geq b$ then $a^n \geq b^n$ ” where the domain consists of all integers. Show that $P(0)$ is true.

Proof by Equivalence

To prove $p \rightarrow q$, show that both $p \rightarrow q$ and $q \rightarrow p$ are true. The validity of this approach is based on the tautology:

$$(p \leftrightarrow q) \leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)]$$

Example 1.29. Prove that if n is a positive integer, then n is odd if and only if n^2 is odd.

Example. Integer n is odd if and only if n^2 is odd.

Proof. **Given:**

- Let p : “ n is odd”
- Let q : “ n^2 is odd”

To prove: $p \leftrightarrow q$ (i.e., n is odd $\iff n^2$ is odd)

Part 1: $p \rightarrow q$ (If n is odd, then n^2 is odd)

Assume n is odd. Then by definition:

$$\begin{aligned} n &= 2k + 1 \quad \text{for some integer } k \\ n^2 &= (2k + 1)^2 \\ &= 4k^2 + 4k + 1 \\ &= 2(2k^2 + 2k) + 1 \end{aligned}$$

Since $2k^2 + 2k$ is an integer, n^2 has the form of an odd number. □

Part 2: $q \rightarrow p$ (If n^2 is odd, then n is odd)

We prove the contrapositive: If n is not odd (i.e., n is even), then n^2 is not odd (i.e., n^2 is even).

Proof. Assume n is even. Then

$$\begin{aligned} n &= 2k \quad \text{for some integer } k \\ n^2 &= (2k)^2 \\ &= 4k^2 \\ &= 2(2k^2) \end{aligned}$$

Since $2k^2$ is an integer, n^2 has the form of an even number.

We have shown both directions:

- $p \rightarrow q$: If n is odd, then n^2 is odd
- $q \rightarrow p$: If n^2 is odd, then n is odd (via contrapositive)

Therefore, n is odd $\iff n^2$ is odd. □

Example. Proof of Equivalent Statements About Integer n . Let

- p_1 : n is even.
- p_2 : $n - 1$ is odd.
- p_3 : n^2 is even.

The statements are equivalent: $p_1 \leftrightarrow p_2 \leftrightarrow p_3$

Proof. We will show the cyclic implications:

$$p_1 \rightarrow p_2, \quad p_2 \rightarrow p_3, \quad \text{and} \quad p_3 \rightarrow p_1$$

1. $p_1 \rightarrow p_2$: If n is even, then $n - 1$ is odd

Proof. Assume n is even. Then:

$$\begin{aligned} n &= 2k \quad \text{for some integer } k \\ n - 1 &= 2k - 1 \\ &= 2(k - 1) + 1 \quad (\text{which has the form of an odd number}) \end{aligned}$$

□

2. $p_2 \rightarrow p_3$: If $n - 1$ is odd, then n^2 is even

Proof. Assume $n - 1$ is odd. Then:

$$\begin{aligned}
 n - 1 &= 2m + 1 \quad \text{for some integer } m \\
 n &= 2m + 2 \\
 &= 2(m + 1) \\
 n^2 &= [2(m + 1)]^2 \\
 &= 4(m + 1)^2 \\
 &= 2[2(m + 1)^2] \quad (\text{clearly even})
 \end{aligned}$$

□

3. $p_3 \rightarrow p_1$: If n^2 is even, then n is even

We prove the contrapositive: If n is odd, then n^2 is odd.

Let n be odd:

$$\begin{aligned}
 n &= 2k + 1 \\
 n^2 &= (2k + 1)^2 \\
 &= 4k^2 + 4k + 1 \\
 &= 2(2k^2 + 2k) + 1 \quad (\text{which is odd})
 \end{aligned}$$

Thus, if n^2 is even, n must be even.

□

General Case for n Equivalent Propositions

For propositions $p_1, p_2, \dots, p_n$ to be equivalent:

$$p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n$$

This can be established by proving the cyclic implications:

$$(p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \dots \wedge (p_n \rightarrow p_1)$$

Conclusion:

We have shown all required implications, therefore:

$$p_1 \leftrightarrow p_2 \leftrightarrow p_3$$

Example

Let n be an integer. The following statements are equivalent:

1. n is even (divisible by 2).
2. n can be written as $n = 2k$ where $k \in \mathbb{Z}$.

3. n^2 is even.

Proof. We prove Statement 1 $\Leftrightarrow$ Statement 2 $\Leftrightarrow$ Statement 3:

- (1 $\Leftrightarrow$ 2) By definition, n is even iff $\exists k \in \mathbb{Z}$ such that $n = 2k$.
- (2 $\Rightarrow$ 3) If $n = 2k$, then $n^2 = 4k^2 = 2(2k^2)$, which is even.
- (3 $\Rightarrow$ 2) Contrapositive: If n is odd ($n = 2k + 1$), then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$ is odd. Thus if n^2 is even, n must be even.

□

2.1.5 Necessary and sufficient conditions

The phrases *necessary* and *sufficient condition*, as used in formal English, correspond exactly to their definition in logic.

Definition 1.30 (17). If p and q are statements:

- p is a **sufficient condition** for q means “if p then q ”, that is, $(p \Rightarrow q)$
- p is a **necessary condition** for q means “if not p then not q ”, that is, $(\sim p \Rightarrow \sim q)$

In other words, to say

p is sufficient condition for q

means that the occurrence of p is *sufficient* to guarantee the occurrence of q . On the other hand, to say

p is necessary condition for q

means that if p does not occur, then q cannot occur either. The occurrence of p is *necessary* to obtain the occurrence of q . Note that because of the equivalence between a statement and its contrapositive,

p is necessary condition for q means **if q then p**

Consequently,

p is necessary and sufficient condition for q means **p if and only if q**

Example 1.31 (Interpreting necessary and sufficient conditions). Consider the statement:

If Suvesh is eligible to vote, then he is at least 18 years old.

The truth of the condition

Suvesh is eligible to vote

is *sufficient* to ensure the truth of the condition

Suvesh is at least 18 years old.

In addition, the truth of the condition

Suvesh is at least 18 years old

is **necessary** for the condition

Suvesh is eligible to vote

to be true. If Suvesh were younger than 18, then he would not be eligible to vote.

Example 1.32 (Converting a sufficient condition to if...then form). Rewrite the following statement in the form **if p then q** :

Robin's birth on Nepal soil is a sufficient condition for him to be a Nepali citizen.

Solution:

If Robin was born on Nepal soil, then he is a Nepali citizen.

Example 1.33 (Converting a necessary condition to if...then form). Use the contrapositive to rewrite the following statement in two ways:

Susan's attaining age 35 is a necessary condition for her being president of Nepal.

Solution:

Version 1:

If Susan has not attained the age of 35, then she cannot be president of Nepal.

Version 2:

If Susan can be president of Nepal, then she has attained the age of 35.

1.4.4 Mathematical Induction

Besides the above three methods, there is also an important approach to prove a theorem about positive integers, and this is the **mathematical induction**. In natural sciences, induction infers something general from a few cases where the statement is known to hold true. If you have only known liars throughout your life, you may infer that the next person you meet is a liar. This is an induction, as you would be making a conjecture based on a finite number of cases known to you. But, in mathematics, a statement involving a natural number might turn out to be erroneous even if it happens to be true for the first ten, or thousand, or even million natural numbers.

For instance, $n^2 - n + 41$ is prime for $n = 1, 2, 3, \dots, 40$, but for $n = 41$, it is 41×41 , and so it is not a prime for $n = 41$. An even more convincing example is the following. The numbers

$$2^{2^0} + 1 = 3, \quad 2^{2^1} + 1 = 5, \quad 2^{2^2} + 1 = 17, \quad 2^{2^3} + 1 = 257, \quad 2^{2^4} + 1 = 65537$$

are all prime numbers and the 17th century mathematician Pierre de Fermat suggested that $2^{2^n} + 1$ must be prime for every positive integer n . However, a century later, another great mathematician Leonard Euler showed that

$$2^{2^5} + 1 = 641 \times 6700417.$$

This tells us that in mathematics, a lot of care is needed to establish an induction procedure, which proves a mathematical theorem for each of an infinite sequence of cases, without exception. So the method of mathematical induction is such a procedure that involves extremely **believable** logic that we accept it as valid reasoning.

Then the axiom we require is called the **principle of mathematical induction** or **just the principle of induction**.

Principle of mathematical induction

Mathematical induction is a proof technique used to establish that a statement $P(n)$ holds for all natural numbers $n \geq n_0$. It consists of two steps:

♣ **Base Case:** Verify that $P(n_0)$ is true (usually for $n_0 = 0$ or $n_0 = 1$).

♣ **Inductive Step:** Assume $P(k)$ is true for some arbitrary integer $k \geq n_0$ (this is called the *inductive hypothesis*). Then, prove that $P(k + 1)$ follows from $P(k)$.

If both steps hold, then by induction, $P(n)$ is true for all $n \geq n_0$.

The induction technique is used to prove a statement which can be put in the form

$$\forall n \geq n_0, p(n)$$

where $n \in \mathbb{Z}^+$ and n_0 is a fixed integer. That is, suppose we wish to show that $p(n)$ is true for all integers $n \geq n_0$.

The principle of induction is the rule of inference for proving that all items in the list $n_0, n_0 + 1, n_0 + 2, \dots$ have the property $p(n)$. The principle has the two forms:

Weak Form:

The weak form has the rule of inference as:

(i) $p(n_0)$ is true	Basic Step
(ii) $(\forall k \geq n_0) [p(k) \text{ true} \implies p(k + 1) \text{ true}]$	Induction Step

$\therefore (\forall n \geq n_0) [p(n) \text{ true}]$	Conclusion
---	-------------------

The above form of mathematical induction using quantifiers is:

$$p(n_0) \wedge [\forall k \geq n_0, p(k) \implies p(k + 1)] \implies \forall n \geq n_0, p(n)$$

is a tautology.

Strong Form:

The strong form has the rule of inference as:

(i) $p(n_0)$ is true	Basic Step
(ii) $(\forall k \geq n_0) [p(n_0) \wedge p(n_0 + 1) \wedge \cdots \wedge p(k) \text{ true}] \implies p(k + 1) \text{ true}$	Induction Step
$\therefore (\forall n \geq n_0) [p(n) \text{ true}]$	Conclusion

The above form of mathematical induction using quantifiers is:

$$p(n_0) \wedge [\forall k \geq n_0, (p(n_0) \wedge \cdots \wedge p(k)) \implies p(k + 1)] \implies \forall n \geq n_0, p(n)$$

is a tautology.

Example 1.34. Use mathematical induction to prove that the following statement holds for all $n \in \mathbb{Z}^+$:

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Proof. Let $p(n)$ be the statement:

$$p(n) : 1^2 + 2^2 + 3^2 + \cdots + n^2 = \frac{n(n+1)(2n+1)}{6}$$

Basic step: We start with the statement $p(1)$:

$$p(1) : 1^2 = \frac{1(1+1)(2 \cdot 1 + 1)}{6} = \frac{6}{6} = 1$$

which is true. So $p(1)$ is true.

Induction step: We assume that the statement is true for some positive integer k , and we then show that it holds for the next positive integer $k + 1$. Assume:

$$1^2 + 2^2 + 3^2 + \cdots + k^2 = \frac{k(k+1)(2k+1)}{6}$$

holds and we then prove that

$$1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 = \frac{(k+1)(k+2)(2k+3)}{6}$$

Now for $p(k+1)$:

$$\begin{aligned} L.H.S &= 1^2 + 2^2 + 3^2 + \cdots + k^2 + (k+1)^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2 \\ &= (k+1) \left[\frac{k(2k+1)}{6} + (k+1) \right] \\ &= \frac{1}{6}(k+1) [2k^2 + k + 6k + 6] \\ &= \frac{1}{6}(k+1)(2k^2 + 7k + 6) \\ &= \frac{1}{6}(k+1)(k+2)(2k+3) \\ &= \frac{(k+1)(k+2)(2k+3)}{6} = R.H.S \end{aligned}$$

Thus $p(k+1)$ is true. Hence by mathematical induction $p(n)$ holds for all $n \in \mathbb{Z}^+$. □

Example 1.35. Prove by mathematical induction that $3 \mid (n^3 - n)$.

Proof. Let $p(n)$ be the statement:

$$p(n) : 3 \mid (n^3 - n)$$

Basic step: We start with $n_0 = 1$ and show that $p(1)$ is true. For $n_0 = 1$:

$$p(1) : 3 \mid (1^3 - 1) = 3 \mid 0$$

So $p(1)$ is true.

Induction step: Suppose that the statement $p(n)$ is true for some integer $k \in \mathbb{Z}^+$. That is:

$$p(k) : 3 \mid (k^3 - k)$$

is true. We shall now prove that:

$$p(k+1) : 3 \mid ((k+1)^3 - (k+1))$$

is true. Now:

$$(k+1)^3 - (k+1) = k^3 + 3k^2 + 3k + 1 - k - 1 = k^3 - k + 3k^2 + 3k$$

Since:

$$3 \mid (k^3 - k) \implies \exists m \in \mathbb{Z}^+ \text{ such that } k^3 - k = 3m$$

Using this relation:

$$(k+1)^3 - (k+1) = 3m + 3(k^2 + k) = 3(m + k^2 + k)$$

Since $m + k^2 + k \in \mathbb{Z}^+$, so $3 \mid ((k+1)^3 - (k+1))$. Therefore $p(k+1)$ is true, and hence by mathematical induction $3 \mid (n^3 - n)$ for all $n \in \mathbb{Z}^+$. □

Theorem 1.36. For every positive integer n ,

$$P(n) : 1 + 2 + 3 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. We proceed by mathematical induction.

Base Case (For $n = 1$):

$$L.H.S. = 1 = \frac{1(1+1)}{2} = 1 = R.H.S.$$

The statement holds.

Inductive Step: Assume the formula holds for $n = k$, i.e.,

$$1 + 2 + \cdots + k = \frac{k(k+1)}{2}. \quad (\text{Inductive Hypothesis})$$

We must show it holds for $n = k + 1$:

$$1 + 2 + \cdots + k + (k+1) = \frac{(k+1)(k+2)}{2}.$$

Starting from the left-hand side (LHS):

$$\begin{aligned}
 \text{LHS} &= (1 + 2 + \cdots + k) + (k + 1) \\
 &= \frac{k(k+1)}{2} + (k+1) \quad (\text{by the inductive hypothesis}). \\
 &= \frac{k(k+1)}{2} + \frac{2(k+1)}{2} \\
 &= \frac{(k+1)(k+2)}{2} \\
 &= \text{RHS}.
 \end{aligned}$$

Thus, the statement holds for $n = k + 1$. By induction, the formula is true for all $n \geq 1$. $\square$

Example 1.37. Recall that $n! = n \cdot (n-1) \cdot (n-2) \cdots 2 \cdot 1$. For any natural number n such that $n \geq 4$, prove that $n! > n^2$.

Solution. Let $n_0 = 4$. Let $T = \{n \in \mathbb{N} : n \geq 4 \text{ and } n! > n^2\}$. Prove by induction for every natural number n that $n \in T$ provided that $n \geq 4$.

(Base step) Show that $4 \in T$. Since $4! = 24$ and $4^2 = 16$, we have $4! > 4^2$, so $4 \in T$.

(Inductive step) Let $n \in T$, then show that $n + 1 \in T$. As before, it is trivial to show that $n + 1 \geq 4$, so it only remains to show that $(n + 1)! > (n + 1)^2$. To prove this, use the following chain of equalities and inequalities:

$$\begin{aligned}
 (n + 1)! &= (n + 1) \cdot n! \quad (\text{definition of } n!) \\
 &> (n + 1) \cdot n^2 \quad (\text{using the inductive hypothesis}) \\
 &> (n + 1) \cdot (n + 1) \quad (\text{use Example 2 in Section 1.7.2 with } n \geq 4 \geq 2) \\
 &= (n + 1)^2
 \end{aligned}$$

Therefore, $n + 1 \in T$.

By the Principle of Mathematical Induction, $T = \{n \in \mathbb{N} : n \geq 4\}$.

1.4.5 Application: Fibonacci Numbers

A famous and often-studied sequence of numbers, called the **Fibonacci numbers**, was defined by Leonardo Fibonacci (1170–1250, born in Italy).¹ The first few numbers in this sequence are

$$1, 1, 2, 3, 5, 8, 13, 21, \dots$$

Denote the n th Fibonacci number by F_n , and let the first element of the sequence be denoted as F_0 . The defining rule for the elements of this sequence is

$$F_0 = F_1 = 1$$

and

$$F_n = F_{n-1} + F_{n-2} \text{ for } n \geq 2.$$

Leonardo developed the sequence in predicting the size of a population of rabbits. F_n is the number of pairs of rabbits he predicted one would have n months after buying a pair of baby rabbits under the assumption that:

¹The Fibonacci numbers were defined by Leonardo of Pisa, *filius* (son of) Bonacci, who lived around 1200.

- A pair of rabbits matured in one month
- Produced a pair of offspring each month thereafter

Month (n)	Old Pairs	New Pairs	Total (F_n)
0	1	0	1
1	1	0	1
2	1	1	2
3	2	1	3
4	3	2	5
5	5	3	8

Furthermore, he assumed that the rabbits always produced a male and a female as each pair of offspring. So:

- $F_0 = 1$ for the pair just purchased
- $F_1 = 1$, because after one month, the original pair has just matured
- $F_2 = 2$, because the original pair has had one pair of offspring
- $F_3 = 3$, because:
 - The original pair has had another pair of offspring
 - The first offspring have just matured

What happens during month n ?

- All rabbits alive during month $n - 1$ are still alive
- All rabbits alive during month $n - 2$ have matured
- Each pair has had one pair of offspring

Hence, the recurrence relation:

$$F_n = F_{n-1} + F_{n-2}$$

The Fibonacci numbers are probably best known as a source of recreational mathematics but are also the source of inspiration for searching and sorting methods. Many results concerning Fibonacci numbers are proved by induction. Example 1.38 shows a typical proof.

Example 1.38. Show that the identity $F_1 + F_3 + F_5 + \cdots + F_{2n-1} = F_{2n} - 1$ is true for all $n \geq 1$.

Solution. Let

$$P(n) = \{n \in \mathbb{N} : n \geq 1 \text{ and } F_1 + F_3 + \cdots + F_{2n-1} = F_{2n} - 1\}$$

We prove that $P(n)$ is true for all $n \geq 1$.

(Base step) Prove the result for $n = 1$. The left-hand side in this case is just the sum of all the Fibonacci numbers starting with F_1 and ending with $F_{2 \cdot 1 - 1}$. There is just one such Fibonacci number, F_1 , and the value of the left-hand side is 1. The right-hand side is $F_{2 \cdot 1} - 1 = F_2 - 1 = 2 - 1 = 1$. So, the two sides are equal, and $1 \in T$.

(Inductive step) Let $n \geq n_0$. Show that if $n \in T$, then $n + 1 \in T$. Since $n \geq 1$, $n + 1 \geq 1$. Assume for n that

$$F_1 + F_3 + \cdots + F_{2n-1} = F_{2n} - 1$$

and prove that

$$F_1 + F_3 + \cdots + F_{2n-1} + F_{2(n+1)-1} = F_{2(n+1)} - 1$$

The required computation is

$$\begin{aligned} F_1 + F_3 + \cdots + F_{2n-1} + F_{2(n+1)-1} &= (F_1 + F_3 + \cdots + F_{2n-1}) + F_{2(n+1)-1} \\ &= F_{2n} - 1 + F_{2(n+1)-1} \quad (\text{using the inductive hypothesis}) \\ &= (F_{2n} + F_{2n+1}) - 1 \quad (\text{rearranging terms}) \\ &= F_{2n+2} - 1 \quad (\text{using the definition of Fibonacci numbers}) \\ &= F_{2(n+1)} - 1 \end{aligned}$$

Therefore, $n + 1 \in T$.

By the Principle of Mathematical Induction, $T = \{n \in \mathbb{N} : n \geq 1\}$.

1.5 Pigeonhole Principle(Without Proof)

The pigeonhole principle is a fundamental concept in combinatorics with wide-ranging applications. Despite its simplicity, it yields powerful results in various mathematical fields. If we put more than n pigeons into n pigeonholes, at least one pigeonhole will house two or more pigeons. This trivial observation is the basis of ingenious combinatorial arguments, and is the subject of this chapter. Let's begin with the various guises of the pigeonhole principle that are encountered in combinatorics.

Theorem 1.39 (Basic Pigeonhole Principle). *If m objects are placed into n containers where $m > n$, then at least one container must contain at least two objects.*

Proof. By contradiction: If each container held at most one object, the total would be $\leq n$ objects, contradicting $m > n$. □

Example 1.40 (Birthday Problem). In any group of 13 people, at least two share a birth month.

Justification: 13 people (objects) placed into 12 months (containers).

Theorem 1.41 (Generalized Pigeonhole Principle). *If m objects are placed into n containers, then at least one container contains at least $\lceil m/n \rceil$ objects.*

Proof. The proof is again by contradiction: If every container contains at most $\lceil m/n \rceil - 1 < m/n$ objects, there are less than $n(m/n) = m$ objects. The more rigorous formulation is: Given two sets A and B , for any function $f : A \rightarrow B$ there exists $b \in B$ such that

$$|\{x \in A : f(x) = b\}| \geq \left\lceil \frac{m}{n} \right\rceil.$$

□

Example 1.42. In a class of 100 students:

- Divided into 12 groups: At least one group has $\lceil 100/12 \rceil = 9$ students
- Divided into 4 groups: At least one group has $\lceil 100/4 \rceil = 25$ students

1.5.1 Dijkstra's Formulation

Theorem 1.43 (Dijkstra's Version). *For any nonempty finite multiset of real numbers, the maximum is at least the average.*

Proof. Let S be a multiset with average μ . If all elements were $< \mu$, their average would be $< \mu$, a contradiction. $\square$

Example 1.44. For the set $\{2, 5, 7, 3\}$:

- Average = $\frac{2+5+7+3}{4} = 4.25$
- Maximum = $7 \geq 4.25$

1.5.2 Application

Example 1.45 (Hair Counting). According to P&G Hair Facts:

- Average hair count: 100,000
- Maximum observed: 150,000 hairs
- Conservative estimate: No San Franciscan has $> 700,000$ hairs
- San Francisco had 776,733 residents.

Proof. By the pigeonhole principle:

- Pigeons: 776,733 residents
- Holes: 700,001 possible hair counts (0 to 700,000)
- Since $776,733 > 700,001$, at least two residents must share the same hair count

$\square$

Theorem 1.46 (Party Acquaintances). *In any gathering of ≥ 6 people, there exist either:*

- *Three mutual acquaintances, or*
- *Three mutual strangers*

Solution: Select arbitrary person a :

- Among remaining 5+ people:
- Case 1: ≥ 3 acquaintances of a
 - Subcase: Any two know each other $\Rightarrow$ triangle with a
 - Else: All strangers $\Rightarrow$ mutual stranger trio
- Case 2: ≥ 3 strangers to a (symmetric argument)

Theorem 1.47 (Colored Plane). *For any red-blue coloring of the infinite plane, there exist two points exactly 1 yard apart with identical colors.*

Solution: Consider an equilateral triangle ABC with side length 1 yard:

- Pigeons: 3 vertices
- Holes: 2 colors
- By PHP, two vertices must share a color

Theorem 1.48 (Coprime Numbers). *Any $n + 1$ numbers selected from $\{1, 2, \dots, 2n\}$ contain a coprime pair.*

Proof: Partition into consecutive pairs:

$$\{1, 2\}, \{3, 4\}, \dots, \{2n - 1, 2n\}$$

- Pigeons: $n + 1$ numbers
- Holes: n pairs
- Some pair $\{k, k + 1\}$ must be fully selected
- Consecutive integers are coprime ($\gcd(k, k + 1) = 1$)

1.5.3 Advanced Applications

Theorem 1.49 (Number Theory). *Any subset of $\{1, 2, \dots, 2n\}$ of size $n + 1$ contains two numbers where one divides the other.*

Theorem 1.50 (Graph Theory). *In any group of 6 people, there are always 3 mutual friends or 3 mutual strangers.*

1.5.4 Exercises

1. Prove that in any set of 51 integers from $\{1, 2, \dots, 100\}$, there must be two that are consecutive.
2. Show that any sequence of $n^2 + 1$ distinct real numbers contains a monotonic subsequence of length $n + 1$.

Sequence and countable sets

A **sequence** is a function from $\mathbb{N}$ or $\mathbb{N} \cup \{0\}$ to a set S .

We use the notation a_n to denote the image of the integer n . We call a_n a *term* of the sequence, denoted by $\{a_n\}$.

Example 1.51. Consider the sequence $\{a_n\}$, where $a_n = \frac{1}{n}$. The list of terms of this sequence, beginning with a_1 , namely

$$a_1, a_2, a_3, \dots$$

starts with

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \dots$$

A set is called **countable** if it is the set corresponding to some sequence. A set that is not countable is called **uncountable**.

- All finite sets are countable.
- However, not all infinite sets are countable.

Example 1.52. The set of integers $\mathbb{Z}$ is countable.

Proof: For $z \in \mathbb{Z}$, let us define $f : \mathbb{Z} \rightarrow \mathbb{N}$ by

$$f(z) = \begin{cases} 2z, & \text{if } z \geq 0 \\ -(2z + 1), & \text{if } z < 0 \end{cases}$$

Clearly, f is one-to-one and onto. Thus, $\mathbb{Z}$ is countable.

Example 7 *The set of rational numbers $\mathbb{Q}$ is countable.*

Example 8 A set $A = \{x \mid x = 4m, m \in \mathbb{Z}\}$ is countable.

Example 9 A set $B = \{x \mid x \text{ is a real number and } 0 < x < 1\}$ is uncountable.

Example 10 *The set of real numbers $\mathbb{R}$ is uncountable.*

To compute the number of elements in the union of two or more overlapping sets, accounting for over-counting due to intersections.

Recursion

Recursive	Explicit
Uses previous terms $a_n = f(a_{n-1})$	Uses position only $a_n = f(n)$
Needs starting point	Standalone formula

Sequences can be defined either recursively or explicitly:

Recursive: $a_1 = 3, a_n = a_{n-1} + 5$ (uses previous terms)

Explicit: $b_n = n^2$ (directly from position)

1.6 Proof Methods

A **proof** is a valid argument that establishes the truth of a mathematical statement, using the hypotheses of the theorem, if any, axioms assumed to be true, and previously proven theorems.

Using these ingredients and rules of inference, the proof establishes the truth of the statement being proved.

We move from formal proofs, as seen in the previous section, to informal proofs, where more than one inference rule may be used at each step, where steps may be skipped, and where axioms and rules of inference used are not explicitly stated.

1.6.1 Some Terminology

Definition 1.53. A precise and unambiguous description of the meaning of a mathematical term.

Theorem 1.54. *A statement that can be shown to be true using vigorous (strong) mathematical reasoning. A statement that can be shown to be true (sometimes referred to as facts or results). Less important theorems are often called propositions.*

Lemma 1.55. *A lemma is a less important theorem, used as an auxiliary result to prove a more important theorem. It is used as an auxiliary result to prove a theorem. It is a stepping stone to prove a theorem.*

Corollary 1.56. *It is a theorem proven as an easy consequence of a theorem.*

Conjecture 1.6.1. A conjecture is a statement that has been proposed as a true statement. If proved later then it can be used as a theorem but it may be false.

Axiom. Axioms (or postulates) are statements that we assume to be true (algebraic axioms specify rules for arithmetic like commutative laws) without proof. These are building blocks to prove theorems.

A **proof** is a valid argument that establishes the truth of a theorem. The statements used in a proof include axioms, hypotheses (or premises), previously proven theorems and definitions. Rules of inferences, together with definition of terms, are used to draw conclusions from other assertions, tying together the steps of a proof.

1.6.2 Three methods of showing statements of the type $p \rightarrow q$ are true:

1. **Direct proofs:** Assume p is true; the last step establishes q is true.
2. **Proof by Contraposition:** Uses a direct proof of the contrapositive of $p \rightarrow q$, which is $\neg q \rightarrow \neg p$. That is, assume $\neg q$ is true; the last step establishes $\neg p$ is true.
3. **Proof by Contradiction:** To prove that P is true, we assume $\neg P$ is true and reach a contradiction, i.e., that $(r \wedge \neg r)$ is true for some proposition r . In particular, to prove $p \rightarrow q$ we assume $p \rightarrow q$ is false, and get as a consequence a contradiction. Assuming that $p \rightarrow q$ is false $\equiv (\neg p \vee q)$ is false $\equiv (p \wedge \neg q)$ is true.

1.6.3 Formal and Informal Proof

A formal direct proof of a conditional statement $p \rightarrow q$ works as follows: assume p is true, build steps using inference rules, with the final step showing that q is true.

In an informal direct proof, we assume that p is true, and use axioms, definitions and previous theorems, together with rules of inference to show that q must be true.

1.6.4 Direct proof

Assume p is true and the last step establishes q is true.

Before we give an example of direct proof, we need a definition.

Definition 1.57. The integer n is **even** if there exists an integer k such that $n = 2k$ and n is **odd** if there exists an integer k such that $n = 2k + 1$.

Theorem 1.58 (Direct proof). *If n is an odd integer, then n^2 is odd.*

Proof. Let n be an odd integer. Then by definition, $n = 2k + 1$ for some integer k . Then

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1.$$

Since $2k^2 + 2k$ is an integer, n^2 is odd. □

Definition 1.59. An integer a is a **perfect square** if there exists an integer b such that $a = b^2$.

Theorem 1.60. *If m and n are both perfect squares, then mn is also a perfect square.*

Proof. Let m and n be perfect squares. Then there exist integers s and t such that $m = s^2$ and $n = t^2$. Then

$$mn = s^2 t^2 = (st)^2.$$

Hence mn is a perfect square. □

1.6.5 Indirect proof

Direct proofs lead from the hypothesis of a theorem to the conclusion. They begin with the premises, continue with a sequence of deductions, and end with the conclusion. We need other methods of proving theorems of the form $\forall x(P(x) \rightarrow Q(x))$. Proofs of theorems of this type that are not direct proofs, i.e., that do not start with the hypothesis and end with the conclusion, are called indirect proofs. Examples of indirect proofs are proof by contraposition.

1.6.6 Proof by contraposition

Theorem 1.61. *If n is an integer and $3n + 2$ is odd, then n is odd.*

Proof. We prove the contrapositive: if n is not odd (i.e., even), then $3n + 2$ is even. Let $n = 2k$. Then

$$3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1),$$

which is even. Thus the contrapositive holds, so the original statement is true. □

Theorem 1.62. *Prove that if $n = ab$ where a and b are positive integers, then $a \leq \sqrt{n}$ or $b \leq \sqrt{n}$.*

Proof. The contrapositive of $(n = ab) \rightarrow (a \leq \sqrt{n} \vee b \leq \sqrt{n})$ is $\neg(a \leq \sqrt{n} \vee b \leq \sqrt{n}) \rightarrow \neg(n = ab)$, i.e., $(a > \sqrt{n} \wedge b > \sqrt{n}) \rightarrow n \neq ab$. If $a > \sqrt{n}$ and $b > \sqrt{n}$, then $ab > \sqrt{n}\sqrt{n} = n$, so $ab \neq n$. Hence the contrapositive is true, and the original statement follows. □

1.6.7 Proof strategy

When we want to prove a statement of the form $\forall x(P(x) \rightarrow Q(x))$, first we try to prove by direct proof. If a direct proof does not seem to go anywhere, try a proof by contraposition.

Definition 1.63. A real number r is **rational** if there exist integers p and q with $q \neq 0$ such that $r = p/q$. A real number that is not rational is called **irrational**.

Theorem 1.64. *The sum of two rational numbers is rational.*

Proof. Let r and s be rational numbers. Then $r = p/q$ and $s = t/u$ where p, q, t, u are integers, $q \neq 0$, $u \neq 0$. Then

$$r + s = \frac{p}{q} + \frac{t}{u} = \frac{pu + qt}{qu},$$

and $qu \neq 0$. Hence $r + s$ is rational. □

Theorem 1.65. *If n is an integer and n^2 is odd, then n is odd.*

Proof. We attempt a direct proof: suppose n^2 is odd, so $n^2 = 2k + 1$. Then $n = \pm\sqrt{2k + 1}$, which is not helpful. Instead, we prove the contrapositive: if n is even, then n^2 is even. Let $n = 2k$. Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is even. Hence the contrapositive holds, so the original statement is true. □

1.6.8 Proof by contradiction

Suppose we want to prove that a statement p is true. Suppose we can find a contradiction q such that $\neg p \rightarrow q$ is true; this can only happen when p is true. For $p \rightarrow q$, we use $(p \wedge \neg q) \rightarrow F$.

Theorem 1.66. *$\sqrt{2}$ is irrational.*

Proof. Assume, for contradiction, that $\sqrt{2}$ is rational. Then $\sqrt{2} = a/b$ where a and b are integers with $b \neq 0$ and no common factors. Squaring both sides gives $2b^2 = a^2$. Thus a^2 is even, so a is even; write $a = 2c$. Then $2b^2 = 4c^2 \Rightarrow b^2 = 2c^2$, so b^2 is even and hence b is even. This contradicts the assumption that a and b have no common factor (both are divisible by 2). Therefore $\sqrt{2}$ is irrational. □

Theorem 1.67. *If $3n + 2$ is odd, then n is odd.*

Proof. Assume, for contradiction, that n is even and $3n + 2$ is odd. Let $n = 2k$. Then $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$, which is even. This contradicts the assumption that $3n + 2$ is odd. Hence n must be odd. □

1.6.9 Proof by equivalence

To prove $p \leftrightarrow q$, show that $p \rightarrow q$ and $q \rightarrow p$ are both true. The validity of this approach is based on the tautology

$$(p \leftrightarrow q) \leftrightarrow [(p \rightarrow q) \wedge (q \rightarrow p)].$$

Example 1.68. Prove that if n is a positive integer, then n is odd if and only if n^2 is odd.

Proof. We have already shown $p \rightarrow q$ (odd $\Rightarrow$ odd square) and $q \rightarrow p$ (square odd $\Rightarrow$ odd) by contraposition. Hence n is odd $\leftrightarrow n^2$ is odd. $\square$

Sometimes a theorem states that several propositions are equivalent. Such a theorem states that propositions $p_1, p_2, \dots, p_n$ are equivalent. This can be written as $p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n$, which states that all propositions have the same truth values. This can be proved by showing

$$(p_1 \leftrightarrow p_2 \leftrightarrow \dots \leftrightarrow p_n) \leftrightarrow (p_1 \rightarrow p_2) \wedge (p_2 \rightarrow p_3) \wedge \dots \wedge (p_n \rightarrow p_1).$$

Example 1.69. Show that these statements about integer n are equivalent:

$$p_1 : n \text{ is even, } p_2 : n - 1 \text{ is odd, } p_3 : n^2 \text{ is even.}$$

Proof. We show $p_1 \rightarrow p_2$, $p_2 \rightarrow p_3$, and $p_3 \rightarrow p_1$.

- $p_1 \rightarrow p_2$: If n is even, $n = 2k$, then $n - 1 = 2k - 1 = 2(k - 1) + 1$, which is odd.
- $p_2 \rightarrow p_3$: If $n - 1$ is odd, then $n - 1 = 2k + 1$, so $n = 2k + 2 = 2(k + 1)$ is even, thus n^2 is even.
- $p_3 \rightarrow p_1$: If n^2 is even, then n must be even (by contrapositive of "if n is odd then n^2 is odd").

Hence all three are equivalent. $\square$

1.6.10 Proof by cases

Sometimes it is difficult to use a single argument that holds for all cases. Proof by cases uses the equivalence:

$$[(p_1 \vee p_2 \vee \dots \vee p_n) \rightarrow q] \equiv [(p_1 \rightarrow q) \wedge (p_2 \rightarrow q) \wedge \dots \wedge (p_n \rightarrow q)].$$

Example 1.70. Prove that if n is an integer then $n^2 \geq n$.

Proof. We split into three cases.

- **Case $n = 0$:** $n^2 = 0 = n$.
- **Case $n \geq 1$:** Multiply both sides of $n \geq 1$ by n (positive) gives $n^2 \geq n$.
- **Case $n \leq -1$:** Then $n^2 \geq 0$ and $0 \geq n$ (since n is negative), so $n^2 \geq n$.

Thus in all cases, $n^2 \geq n$. $\square$

1.6.11 Exhaustive proof

This is a special form of proof by cases, when there is a finite and small number of examples for which we need to prove a fact. Such proofs proceed by exhausting all possibilities.

Example 1.71. Prove that $(n + 1)^2 \geq 3^n$ if n is a positive integer with $n \leq 2$.

Proof. We examine the cases $n = 1, 2$:

- $n = 1$: $(1 + 1)^2 = 4 \geq 3^1 = 3$.

- $n = 2$: $(2 + 1)^2 = 9 \geq 3^2 = 9$.

Hence the inequality holds. □

Example 1.72. Prove that $(n + 1)^2 \geq 3^n$ if n is a positive integer with $n \leq 4$.

Proof. Check $n = 1, 2, 3, 4$:

- $n = 1$: $4 \geq 3$.
 - $n = 2$: $9 \geq 9$.
 - $n = 3$: $(3 + 1)^2 = 16 \geq 27$? Actually $16 \geq 27$ is false. The statement is false for $n = 3$. (The original document had an error; we note it.)
-

1.6.12 Existence Proofs

Existence proofs prove statements of the form $\exists x P(x)$.

- **Constructive existence proof:** find a such that $P(a)$ is true.

Example 1.73. Show that there is a positive integer that can be written as a sum of cubes of positive integers in two different ways.

Proof. $1729 = 10^3 + 9^3$ and also $1729 = 12^3 + 1^3$. □

- **Non-constructive existence proof:** show that $\exists x P(x)$ without explicitly giving an a for which $P(a)$ is true.

Example 1.74. Show that there exist irrational numbers x and y such that x^y is rational.

Proof. We know $\sqrt{2}$ is irrational. Consider $\sqrt{2}^{\sqrt{2}}$. There are two cases:

- If $\sqrt{2}^{\sqrt{2}}$ is rational, take $x = \sqrt{2}$, $y = \sqrt{2}$.
- If $\sqrt{2}^{\sqrt{2}}$ is irrational, take $x = \sqrt{2}^{\sqrt{2}}$, $y = \sqrt{2}$. Then

$$x^y = (\sqrt{2}^{\sqrt{2}})^{\sqrt{2}} = \sqrt{2}^2 = 2,$$

which is rational.

Thus in either case, such irrationals exist, but we do not know which pair works. □

1.6.13 Mistakes in Proofs

What is the problem with the following proof that $1 = 2$?

1. $a = b$ (given)
2. $a^2 = ab$ (multiply both sides by a)
3. $a^2 - b^2 = ab - b^2$ (subtract b^2)
4. $(a - b)(a + b) = b(a - b)$ (factor)
5. $a + b = b$ (divide by $a - b$)
6. $2b = b$ (since $a = b$)
7. $2 = 1$ (divide by b)

The error is in step 5: dividing by $a - b$ is invalid because $a - b = 0$ (division by zero).

1.7 Mathematical Induction

1.7.1 Principle of Mathematical Induction

To prove that $P(n)$ is true for all positive integers n (where $P(n)$ is a propositional function), we do two steps:

- **Basic step:** verify that $P(0)$ or $P(1)$ is true.
- **Inductive hypothesis:** assume that $P(k)$ is true for an arbitrary k .
- **Inductive step:** prove that $P(k+1)$ is true, i.e., show $P(k) \rightarrow P(k+1)$ for all positive integers k .

As a rule of inference:

$$[P(1) \wedge \forall k(P(k) \rightarrow P(k+1))] \rightarrow \forall n P(n).$$

Example 1.75. Show that if n is a positive integer then

$$1 + 2 + \cdots + n = \frac{n(n+1)}{2}.$$

Proof. Let $P(n)$ be the proposition that the sum of the first n positive integers is $n(n+1)/2$.

- **Basis step:** $P(1)$ is true because $1 = 1 \cdot 2/2 = 1$.
- **Inductive step:** Assume $P(k)$ is true for an arbitrary positive integer k , i.e., $1 + 2 + \cdots + k = k(k+1)/2$. Then

$$\begin{aligned} 1 + 2 + \cdots + k + (k+1) &= \frac{k(k+1)}{2} + (k+1) \\ &= \frac{k(k+1) + 2(k+1)}{2} \\ &= \frac{(k+1)(k+2)}{2}. \end{aligned}$$

Thus $P(k+1)$ is true. By the principle of mathematical induction, $P(n)$ holds for all positive integers n .

□

Example 1.76. Conjecture a formula for the sum of the first n positive odd integers, and prove it by induction.

Proof. The sums for $n = 1, 2, 3, 4, 5$ are:

$$1 = 1, \quad 1 + 3 = 4, \quad 1 + 3 + 5 = 9, \quad 1 + 3 + 5 + 7 = 16, \quad 1 + 3 + 5 + 7 + 9 = 25.$$

We conjecture that $1 + 3 + 5 + \cdots + (2n - 1) = n^2$. Let $P(n)$ be this proposition.

- **Basis step:** $P(1)$: $1 = 1^2$.
- **Inductive step:** Assume $P(k)$: $1 + 3 + \cdots + (2k - 1) = k^2$. Then

$$\begin{aligned} 1 + 3 + \cdots + (2k - 1) + (2(k + 1) - 1) &= k^2 + (2k + 1) \\ &= k^2 + 2k + 1 = (k + 1)^2. \end{aligned}$$

Hence $P(k + 1)$ holds. By induction, the formula holds for all positive integers n .

□

Example 1.77. Show that $1 + 2 + 2^2 + \cdots + 2^n = 2^{n+1} - 1$ for all non-negative integers.

Proof. Let $P(n)$ be the proposition. Basis step: $n = 0$, LHS = $2^0 = 1$, RHS = $2^1 - 1 = 1$. Inductive step: assume $P(k)$ holds. Then

$$1 + 2 + 2^2 + \cdots + 2^k + 2^{k+1} = (2^{k+1} - 1) + 2^{k+1} = 2 \cdot 2^{k+1} - 1 = 2^{k+2} - 1.$$

Thus $P(k + 1)$ holds. By induction, true for all $n \geq 0$.

□

Example 1.78. Prove $n < 2^n$ for all positive integers n .

Proof. Base: $n = 1$, $1 < 2^1 = 2$. Inductive step: assume $k < 2^k$. Then

$$k + 1 < 2^k + 1 < 2^k + 2^k = 2 \cdot 2^k = 2^{k+1},$$

since $1 < 2^k$ for $k \geq 1$. Hence $k + 1 < 2^{k+1}$. By induction, true for all $n \geq 1$.

□

Example 1.79. Prove that $2^n < n!$ for every positive integer n with $n \geq 4$.

Proof. Base: $n = 4$, $2^4 = 16 < 24 = 4!$. Inductive step: assume $2^k < k!$ for some $k \geq 4$. Then

$$2^{k+1} = 2 \cdot 2^k < 2 \cdot k! < (k + 1) \cdot k! = (k + 1)!,$$

since $2 < k + 1$ for $k \geq 4$. Hence $2^{k+1} < (k + 1)!$. By induction, true for all $n \geq 4$.

□

Example 1.80. Harmonic numbers: $H_j = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{j}$. Show that $H_{2^n} \geq 1 + \frac{n}{2}$ for non-negative integers n .

Proof. Base: $n = 0$, $H_{2^0} = H_1 = 1 \geq 1 + 0/2 = 1$. Inductive step: assume $H_{2^k} \geq 1 + k/2$. Then

$$H_{2^{k+1}} = H_{2^k} + \sum_{i=2^k+1}^{2^{k+1}} \frac{1}{i} \geq H_{2^k} + 2^k \cdot \frac{1}{2^{k+1}} = H_{2^k} + \frac{1}{2} \geq (1 + \frac{k}{2}) + \frac{1}{2} = 1 + \frac{k+1}{2}.$$

Thus holds for $k + 1$. By induction, true for all n .

□

Example 1.81. Prove that $n^3 - n$ is divisible by 3 for all positive integers n .

Proof. Base: $n = 1$, $1^3 - 1 = 0$ divisible by 3. Inductive step: assume $k^3 - k$ divisible by 3. Then

$$(k+1)^3 - (k+1) = (k^3 + 3k^2 + 3k + 1) - k - 1 = (k^3 - k) + 3(k^2 + k).$$

Both terms are divisible by 3, so the sum is divisible by 3. Hence true for all n . $\square$

Example 1.82. If S is a finite set with n elements, then S has 2^n subsets.

Proof. Base: $n = 0$, empty set has $2^0 = 1$ subset (itself). Inductive step: assume any set with k elements has 2^k subsets. Let T have $k+1$ elements, pick $a \in T$ and let $S = T \setminus \{a\}$. Then each subset X of S gives two subsets of T : X and $X \cup \{a\}$. There are 2^k subsets of S , so T has $2 \cdot 2^k = 2^{k+1}$ subsets. By induction, true for all n . $\square$

1.8 Summarization of proof techniques

S.N.	Proof Technique	Approach to prove $p \rightarrow q$
1	Exhaustive proof	Demonstrate $p \rightarrow q$ for all possibilities (finite cases)
2	Direct proof	Assume p , deduce q
3	Proof by contraposition	Assume $\neg q$, derive $\neg p$
4	Proof by contradiction	Assume $p \wedge \neg q$, deduce a contradiction
5	Vacuous proof	Show p is false
6	Trivial proof	Show q is true
7	Proof by cases	Demonstrate $p \rightarrow q$ for all cases
8	Proof by equivalence	Show $p \leftrightarrow q$ by proving $p \rightarrow q$ and $q \rightarrow p$

1.9 Valid, Satisfiable, Unsatisfiable Formulas

Definition 1.83. A **valid formula** (tautology) is true under all valuations. Example: $p \rightarrow p$.

Definition 1.84. A **satisfiable formula** can be made true under at least one valuation. Example: $p \vee q$ (true when $p = T, q = F$). Every valid formula is satisfiable.

Definition 1.85. An **unsatisfiable formula** is false under all valuations.

1.9.1 Soundness and Completeness of a Logical System

The axioms are all valid and each inference rule has the property: if both premises are valid then the conclusion is valid. Only valid formulae can be proved in a sound logical system.

Definition 1.86. If ϕ can be proved in a sound logical system, then ϕ is valid. Such a ϕ is called a theorem.

Soundness: If ϕ is a theorem (can be proved), then ϕ is valid.

Completeness: If ϕ is valid, then ϕ is a theorem (can be proved).

1.10 Tableau Method

Why tableau? The main problem with axiomatic systems is that it is hard to construct a proof; we need intuition. Tableau is a method to prove satisfiability of a set of formulas. It is easy to use and suitable for computer implementation. It breaks each formula into components, creating a tree structure called a tableau.

1.10.1 Parsing

Parsing recovers the structure of a formula. Example: $\phi = (p \wedge q) \vee (\neg p \wedge \neg q)$ can be parsed into a tree.

1.10.2 Tableau Construction Rules

If a formula ϕ is not a literal, it has one of the following forms. We apply rules to decompose it.

ϕ	Rule
$\neg\neg B$	Add B
$B \wedge C$	Add B, C
$\neg(B \wedge C)$	Branch: $\neg B \mid \neg C$
$B \vee C$	Branch: $B \mid C$
$\neg(B \vee C)$	Add $\neg B, \neg C$
$B \rightarrow C$	Branch: $\neg B \mid C$
$\neg(B \rightarrow C)$	Add $B, \neg C$
$B \leftrightarrow C$	Branch: $(B \wedge C) \mid (\neg B \wedge \neg C)$
$\neg(B \leftrightarrow C)$	Branch: $(B \wedge \neg C) \mid (\neg B \wedge C)$

1.10.3 Tableau Procedure

1. List the formulas $\phi_1, \phi_2, \dots, \phi_m$.
2. Apply derivation rules to decompose formulas.
3. For each branch, check for contradictions (a formula and its negation).
4. If all branches are closed (contain a contradiction), the original set is unsatisfiable.
5. If a branch remains open and all non-literals are decomposed, the set is satisfiable. A valuation can be read from the open branch: assign T to literals that appear, F to their negations, and arbitrary for missing variables.

Example 1.87. Consider the set $\phi = \{(P \wedge \neg Q) \rightarrow S, \neg Q \vee \neg R, P \wedge T\}$. A tableau for ϕ :

$$\begin{array}{c}
(P \wedge \neg Q) \rightarrow S \\
\neg Q \vee \neg R \\
P \wedge T \\
\hline
P, T \\
\hline
\neg Q \vee \neg R \\
\hline
\neg Q \quad | \quad \neg R \\
\hline
(P \wedge \neg Q) \rightarrow S \\
\hline
\neg(P \wedge \neg Q) \quad | \quad S \\
\hline
\neg P \vee Q \quad | \quad S \\
\hline
\text{close?}
\end{array}$$

An open branch yields a valuation: $V(P) = T$, $V(T) = T$, $V(R) = F$, $V(S) = T$ or F , $V(Q) = T$ (for the rightmost branch). Then check that all formulas are satisfied.

1.10.4 Soundness and Completeness of Propositional Tableau

Theorem 1.88 (Soundness). *If you can construct a closed tableau for $\neg\phi$, then ϕ is valid.*

Theorem 1.89 (Completeness). *If ϕ is valid, then you can construct a closed tableau for $\neg\phi$.*

Corollary 1.90. *If a completed tableau for a formula ϕ is open, then ϕ is satisfiable. If it is closed, then ϕ is unsatisfiable.*

1.10.5 Valid Arguments

An argument with premises $A_1, A_2, \dots, A_n$ and conclusion B is valid if B is a tautological consequence of the premises, i.e., $\{A_1, \dots, A_n\} \models B$. Equivalently, the set $\{A_1, \dots, A_n, \neg B\}$ is unsatisfiable.

Example 1.91. Check validity of:

$$P \rightarrow (Q \wedge R), Q \rightarrow S, R \rightarrow A, \neg P \rightarrow B, \therefore (S \wedge A) \wedge B.$$

We construct a tableau for premises plus negation of conclusion. If all branches close, the argument is valid.